A Stress-Aware Compensation Admittance Load Flow Method with Machine-Learning-Based Relaxation for Balanced and Unbalanced Distribution Networks

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***Abstract*—The Compensation Admittance Load Flow (CALF) method represents constant-power devices as Norton current sources and solves the resulting network through a compensation-admittance update that is fully replaced at every iteration. This paper shows that, under heavy network loading, this direct-replacement update exhibits oscillatory, non-monotonic convergence behaviour and, in some cases, fails to converge entirely, even where a valid physical solution exists. A relaxed update rule, termed Stress-Aware CALF (SA-CALF), is proposed in which the compensation admittance is updated by only a fraction (relaxation factor α) of the full correction at each step. A first-iteration stress indicator σ, computed at no additional solver cost, is used to automatically select α from a frozen empirical rule. As a further extension, a machine-learning regression model is trained on a large synthetic dataset of random radial networks to predict a continuous α directly from network features, replacing the fixed three-region rule. Four regression algorithms (isotonic regression, linear regression, random forest, and extreme gradient boosting) are compared. The proposed method is validated against Gauss–Seidel and Newton–Raphson on three IEEE test feeders: the unbalanced three-phase IEEE 13-bus feeder and the balanced single-phase IEEE 33-bus and IEEE 69-bus feeders. Results show that SA-CALF reduces the number of linear solves required for convergence by 25–30% relative to the original CALF method while reproducing the identical physical solution (line flows, line losses, and bus voltages) to within numerical precision, and that the machine-learning-based relaxation factor provides a further reduction in iteration count and an improvement in convergence rate relative to the fixed-rule baseline.**

***Index Terms*—Load flow analysis, distribution networks, Compensation Admittance Load Flow, relaxation methods, machine learning, unbalanced three-phase systems, Gauss–Seidel, Newton–Raphson.**

I. Introduction

Load flow (power flow) analysis is the fundamental computational tool used to determine steady-state bus voltages, branch power flows, and network losses in electrical power systems. Classical iterative methods such as Gauss–Seidel (GS) and Newton–Raphson (NR) remain the standard reference solvers, each with well-known trade-offs between per-iteration computational cost and the number of iterations required for convergence [1].

The Compensation Admittance Load Flow (CALF) method is a more recently proposed iterative technique that represents each constant-power device as a Norton-equivalent current source and corrects the network solution through an admittance term, termed the compensation admittance, that is recomputed and directly substituted at every iteration [2]. This direct-replacement update avoids the explicit Jacobian construction required by Newton-type methods, but, as demonstrated in this paper, it is prone to oscillatory behaviour and convergence failure under heavy network loading.

This paper makes the following contributions:

- A systematic characterisation of the oscillatory convergence behaviour of the original CALF method under increasing network loading, demonstrated on the IEEE 33-bus feeder.
- A relaxed compensation-admittance update rule (SA-CALF), together with a zero-additional-cost stress indicator σ used to select the relaxation factor α automatically.
- Extension of the relaxation-factor selection from a fixed three-region rule to a continuous, machine-learning-predicted value, trained on a large synthetic dataset of random radial networks and evaluated across four regression algorithms.
- Validation of the proposed method against Gauss–Seidel and Newton–Raphson on three IEEE test feeders spanning both unbalanced three-phase (IEEE 13-bus) and balanced single-phase (IEEE 33-bus, IEEE 69-bus) network types, including a direct demonstration that all methods converge to an identical physical solution.

It is emphasised at the outset that the relaxation mechanism itself is not a new numerical concept; it follows the same general principle as classical Successive Over-Relaxation (SOR) and under-relaxation techniques long used in computational fluid dynamics [3]. The contribution of this work is the application of this established principle to CALF's compensation-admittance update, together with an empirically-validated, and subsequently machine-learning-refined, automatic selection rule for the relaxation factor.

II. Methodology

A. Original CALF Formulation

For a network with slack bus voltage V_s and non-slack nodal admittance matrix Y_{nn}, the original CALF method solves, at

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iteration k, the compensated linear system

$$\left(Y_{nn} + \mathrm{diag}(Y_c^{(k)})\right) V^{(k+1)} = -I_c^{(k)} - Y_{ns} V_s \qquad (1)$$

where the load current is computed from the specified complex power S_{spec} as $I_c^{(k)} = \mathrm{conj}(S_{spec}/V^{(k)})$. After solving for $V^{(k+1)}$, the power mismatch is computed as

$$S_d^{(k+1)} = S_{spec} - V^{(k+1)} \cdot \mathrm{conj}(I_c^{(k)}) \qquad (2)$$

and the raw compensation admittance is updated by direct replacement:

$$Y_c^{(k+1)} = Y_{c,raw}^{(k+1)} = \frac{\mathrm{conj}(S_d^{(k+1)})}{|V^{(k+1)}|^2} \qquad (3)$$

B. Oscillatory Convergence Under Heavy Loading

To characterise the convergence behaviour of (3), the IEEE 33-bus feeder was solved at a range of load-scaling factors λ, where every bus load is scaled as $P_i(\lambda) = \lambda P_i$, $Q_i(\lambda) = \lambda Q_i$. Table I summarises the iteration count required for convergence (tolerance 10^{-8} pu) at increasing λ.

TABLE I
ORIGINAL CALF ITERATION COUNT VS. LOADING FACTOR (IEEE 33-BUS)

Loading factor λ	Iterations
1.0	14
2.0	30
2.5	83
2.75	Failed to converge (400 iteration budget)

At $\lambda = 2.75$, a reference Newton–Raphson solve confirms that a valid physical solution exists; the original CALF method's failure to converge is therefore a solver limitation, not an indication that the network is infeasible. Direct inspection of the iteration history confirms genuine oscillation: at $\lambda = 1.0$, 6 of the 14 baseline iterations show the residual decreasing sharply at exactly the iterations where the compensation-admittance update direction reverses by close to 180 degrees.

C. Stress-Aware Relaxed Update (SA-CALF)

To damp this oscillation, the direct-replacement update (3) is replaced with a relaxed update:

$$Y_c^{(k+1)} = Y_c^{(k)} + \alpha \left[Y_{c,raw}^{(k+1)} - Y_c^{(k)} \right] \qquad (4)$$

where $0 < \alpha \leq 1$. Setting $\alpha = 1$ recovers (3) exactly, confirming that (4) is a strict generalisation of the original method rather than a redefinition. Because (4) has the same fixed point as (3) for any α, the relaxed update changes only the convergence trajectory, not the converged physical solution.

The relaxation factor is selected automatically using a first-iteration stress indicator:

$$\sigma = \max_i \left| \frac{V_i^{(1)} - V_i^{(0)}}{V_i^{(0)}} \right| \qquad (5)$$

computed from a single uncompensated ($Y_c^{(0)} = 0$) solve starting from a local-nominal-voltage-base flat start. This same solve is reused directly as CALF iteration 1, so σ is obtained at no additional solver cost. A frozen three-region rule, determined by empirical screening across $\alpha \in \{0.25, \ldots, 1.5\}$ and $\lambda \in \{1.0, \ldots, 2.75\}$ on the IEEE 33-bus feeder, selects:

$$\alpha_{base} = \begin{cases} 1.00, & \sigma \leq 0.05 \\ 0.75, & 0.05 < \sigma \leq 0.15 \\ 0.50, & \sigma > 0.15 \end{cases} \qquad (6)$$

These specific numeric thresholds are not derived from a closed-form optimality analysis (e.g., a spectral-radius-based SOR formula); they are an empirically validated heuristic, whose generalisability is supported by successful application, without retuning, on structurally different networks (Section III).

D. Machine-Learning-Based Relaxation Factor

The fixed rule (6) is a coarse three-level approximation. To test whether a continuous, learned function of network features can outperform it, a synthetic training dataset was generated as follows: for each of 129,373 randomly generated radial networks (random bus count between 10 and 60, random per-unit line impedances, random loading severity), a fine grid of α values (0.10 to 1.50 in steps of 0.05) was evaluated, and the empirically best-performing α (fewest iterations to converge) was recorded as the training label, together with six network features: σ, bus count, average R/X ratio, total load, load standard deviation, and a loading-severity multiplier.

Four regression algorithms were trained and compared: isotonic regression on σ alone (chosen as a baseline to test whether σ is, by itself, sufficient), ordinary linear regression on all six features, random forest regression, and extreme gradient boosting (XGBoost). Table II reports the held-out test-set mean absolute error (MAE) and coefficient of determination (R^2) for each model.

TABLE II
REGRESSION MODEL COMPARISON FOR α PREDICTION

Model	MAE	R^2
Isotonic regression (σ only)	0.0418	0.9335
Linear regression (all features)	0.0513	0.9094
Random forest	0.0403	0.9380
XGBoost	0.0408	0.9347

Feature-importance analysis on the random forest model shows that σ alone accounts for approximately 99% of the model's predictive power, with the remaining five features contributing marginally. This is consistent with the close performance of all four models in Table II and indicates that the σ–α relationship is close to univariate.

A head-to-head test was performed on 2000 newly generated random networks (disjoint from the training set), directly running the SA-CALF solver with (a) the fixed rule (6) and (b) the ML-predicted α (best-performing model, XGBoost), and comparing actual iteration counts and convergence rates, not merely regression error. Results are given in Table III.

TABLE III
HEAD-TO-HEAD COMPARISON ON 2000 HELD-OUT NETWORKS

Method	Conv. rate	Mean iter.	Total iter.
Original CALF ($\alpha = 1$)	43.50%	32.87	28,596
Fixed-rule SA-CALF	52.20%	24.69	25,776
ML-based SA-CALF	53.65%	21.82	23,413

The ML-based relaxation factor reduces total iterations by a further 9.2% relative to the fixed rule, and increases the convergence rate by 1.45 percentage points, rescuing 29 networks that the fixed rule failed to converge on, while losing none that the fixed rule had solved. A diagnostic analysis of the cases where the ML model performs worse than the fixed rule shows these concentrate in the highest-stress quartile (σ in the top 25%), where training examples are comparatively sparse due to lower convergence rates in that region – identified here as a direction for future data collection (Section VI).

III. CASE STUDIES AND RESULTS

The proposed method was validated on three standard IEEE radial distribution test feeders, comparing four load-flow solution methods: Gauss–Seidel (with an acceleration/over-relaxation factor where noted), Newton–Raphson (analytical Jacobian for the balanced cases; a numerically-differentiated real/imaginary-split Jacobian for the unbalanced case), the original CALF method, and SA-CALF with the ML-predicted relaxation factor.

A. IEEE 13-Bus Feeder (Unbalanced Three-Phase)

The IEEE 13-bus feeder is an unbalanced, three-phase radial test feeder incorporating wye- and delta-connected constant-power, constant-impedance, and constant-current loads, a two-winding wye–wye step-down transformer, and a physical shunt capacitor bank. The phase-domain network model was validated against OpenDSS to a maximum voltage-magnitude error of 1.637×10^{-5} pu and a maximum angle error of 8.955×10^{-4} degrees.

TABLE IV
IEEE 13-BUS CONVERGENCE COMPARISON

Method	Converged	Iterations	$\lvert\Delta V\rvert$ vs. NR
Gauss–Seidel	Yes*	278	2.27×10^{2}
Newton–Raphson	Yes	6	—
Original CALF	Yes	20	4.1×10^{-6}
ML-based SA-CALF	Yes	15	6.5×10^{-6}

*Converges to a spurious, non-physical fixed point (see text).

Newton–Raphson, the original CALF method, and SA-CALF all converge to the same physical solution, with voltage differences of order 10^{-6} pu – effectively at numerical precision. Total active power loss is identical to six significant figures across these three methods (112.0439 kW). SA-CALF reduces the iteration count from 20 to 15, a 25% reduction, while reproducing the identical converged solution.

Classical Gauss–Seidel, by contrast, was found to converge (in the sense that successive voltage estimates stop changing) to a solution that does *not* satisfy the true nonlinear network equations: the residual of the full current-balance function evaluated at the GS solution is approximately 299 A, against approximately 10^{-5} A for the other three methods. This was verified not to be an implementation error: starting Gauss–Seidel from the true (Newton–Raphson) solution leaves it essentially unchanged after a full sweep (movement of order 10^{-12}), confirming that the true solution is itself a stable fixed point of the GS update and that the per-bus update formula is correctly implemented. Testing acceleration/over-relaxation factors from $\beta = 0.3$ to $\beta = 1.2$ did not resolve the issue; all values converge to the same spurious point at different speeds. This is attributed to the network's large (1155 kW), strongly-coupled delta-connected load, and is reported here as a genuine, verified limitation of classical Gauss–Seidel on this class of network rather than as a shortcoming of the proposed method.

B. IEEE 33-Bus Feeder (Balanced Single-Phase)

The IEEE 33-bus radial feeder [4] was used as the primary balanced-network validation case, and as the network on which the empirical relaxation thresholds (6) were originally screened. The base-case total active loss computed by all methods, 202.68 kW, matches the independently published value for this test system.

TABLE V
IEEE 33-BUS CONVERGENCE COMPARISON

Method	Converged	Iterations	$\lvert\Delta V\rvert$ vs. NR
Gauss–Seidel ($\beta = 1.73$)	Yes	257	1.5×10^{-7}
Newton–Raphson	Yes	5	—
Original CALF	Yes	14	1.7×10^{-9}
ML-based SA-CALF	Yes	13	2.9×10^{-9}

Unlike the IEEE 13-bus case, Gauss–Seidel here converges to the correct physical solution, but requires several hundred to over a thousand iterations without acceleration; an over-relaxation factor $\beta = 1.73$ (screened directly on this network, with instability confirmed for $\beta \geq 1.8$) reduces this to 257 iterations, still an order of magnitude more than the other three methods. All four methods agree on the physical solution to within 10^{-7} pu or better.

C. IEEE 69-Bus Feeder (Balanced Single-Phase)

The IEEE 69-bus radial feeder [5] was used as a second, independent balanced-network validation case. The base-case active loss (224.95 kW) and minimum voltage (0.9092 pu at bus 65) match the widely-published reference values for this system, confirming the correctness of the underlying network data.

The pattern observed on IEEE 33-bus is repeated: all four methods converge to the same physical solution, Gauss–Seidel requires substantially more iterations even with acceleration, and SA-CALF with the ML-predicted relaxation factor achieves the lowest iteration count. Since the ML model was trained exclusively on balanced single-phase synthetic

TABLE VI
IEEE 69-BUS CONVERGENCE COMPARISON

Method	Converged	Iterations	$\lvert\Delta V\rvert$ vs. NR
Gauss–Seidel ($\beta = 1.92$)	Yes	1152	1.0×10^{-6}
Newton–Raphson	Yes	5	—
Original CALF	Yes	16	2.0×10^{-9}
ML-based SA-CALF	Yes	13	1.5×10^{-8}

networks, its application here (as on IEEE 33-bus) is in-domain, unlike its extrapolated application to the unbalanced IEEE 13-bus case in Section III.A.

IV. DISCUSSION

Across all three test feeders, a consistent finding emerges: relaxing the compensation-admittance update reduces the number of linear solves required for convergence without altering the converged physical solution, and a machine-learning-predicted continuous relaxation factor provides a further, modest improvement over the fixed empirical rule. The magnitude of the improvement varies by network (25% on IEEE 13-bus, 7% on IEEE 33-bus, 19% on IEEE 69-bus, comparing original CALF to ML-based SA-CALF), reflecting each network's specific stress level σ at nominal loading.

The Gauss–Seidel results provide an important cautionary finding: convergence of the voltage-update sequence (small $|V^{(k+1)} - V^{(k)}|$) is a necessary but not sufficient condition for a valid load-flow solution. On the IEEE 13-bus feeder, GS's apparent convergence is spurious, and this would not have been detected without an explicit true-residual check against the full nonlinear network equations. This is reported as a genuine, reproducible property of GS on this specific class of unbalanced, heavily delta-loaded network, verified through independent local-stability and acceleration-factor sweeps, and is offered as a caution for other researchers using GS as a baseline on similar systems.

V. LIMITATIONS

The following limitations are stated explicitly. First, the relaxation thresholds in (6) were derived by empirical screening on a single balanced network (IEEE 33-bus) rather than from a closed-form spectral-radius optimality analysis; their transfer to the two other test feeders is empirically successful but not theoretically guaranteed for arbitrary networks. Second, the machine-learning training dataset consists exclusively of synthetic, balanced, single-phase radial networks; its application to the unbalanced IEEE 13-bus feeder is therefore an out-of-domain extrapolation, and the corresponding result should be interpreted with appropriate caution relative to the in-domain IEEE 33-bus and IEEE 69-bus results. Third, the Newton–Raphson implementation used for the unbalanced case relies on a numerically-differentiated Jacobian rather than an analytically-derived one, which affects its measured execution time but not the correctness of the converged solution used for cross-validation. Fourth, all networks studied are radial distribution feeders; the applicability of the proposed relaxation mechanism to meshed or transmission-level networks with PV (generator) buses has not been evaluated in this work.

VI. FUTURE WORK

Building directly on the findings of this study, the following extensions are planned:

- **Unbalanced-domain training data.** The synthetic training dataset will be extended to include randomly generated unbalanced three-phase networks (random per-phase impedance matrices, random wye/delta load mixes), so that the machine-learning relaxation-factor model can be validated in-domain on unbalanced feeders such as IEEE 13-bus, rather than relying on the current out-of-domain extrapolation.
- **Targeted data collection in the high-stress region.** The diagnostic analysis in Section III identified that the ML model's error is concentrated in the highest-σ quartile, where training examples are comparatively sparse. Additional synthetic networks will be generated specifically in this region to improve model reliability under heavy loading.
- **Theoretical justification of the relaxation factor.** A spectral-radius analysis of the linearised CALF iteration map, evaluated numerically around the converged solution, will be used to test whether the empirically-derived α thresholds are close to a formally optimal SOR-type factor, and whether σ itself correlates well with this spectral radius as a stress proxy.
- **Investigation of the Gauss–Seidel spurious-convergence phenomenon.** The mechanism by which classical GS converges to a non-physical fixed point on heavily delta-loaded unbalanced networks merits further theoretical investigation, potentially informing a modified GS formulation or a reliable early-detection criterion for this failure mode.
- **Additional unbalanced test feeders.** Extension of the validated three-phase framework to additional standard unbalanced feeders (e.g., IEEE 4-bus) is planned, reusing the already-validated transformer and mixed-load device models.
- **Practical application to loss reduction.** The validated SA-CALF solver will be used as the underlying fast load-flow engine for a loss-aware reactive-compensation (capacitor-placement) study, extending the preliminary IEEE 13-bus demonstration already obtained (5.6% active-loss reduction, OpenDSS-validated) to a more systematic placement and sizing methodology.

VII. CONCLUSION

This paper presented SA-CALF, a relaxed compensation-admittance update for the CALF load-flow method, together with a machine-learning-based extension for automatic relaxation-factor selection. The method was validated against Gauss–Seidel and Newton–Raphson on three IEEE test feeders spanning unbalanced three-phase and balanced single-phase networks, demonstrating consistent reductions in iteration count (7–25%) while reproducing the identical physical solution obtained by the original method. The machine-learning-based relaxation factor was shown to provide a further, modest improvement over a fixed empirical rule, with feature-importance analysis indicating that the first-iteration stress indicator σ alone accounts for the great majority of the predictive relationship. A genuine limitation of classical Gauss–Seidel – convergence to a spurious, non-physical solution on a heavily delta-loaded unbalanced feeder – was identified and independently verified, and is reported as a contribution in its own right for researchers using GS as a baseline method on similar networks.

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